Diffusion, Gelation, Spherification

1. How many seconds would it take for a molecule with a diffusion constant of $3 \cdot 10^{-6}$ cm^2/s to diffuse 1 mm?

L=sqrt4Dt

$t = L^2/4D = (0.1\ cm)^2/4*3\cdot10^{-6}\ cm^2/s = 833\ sec = 13\ min$

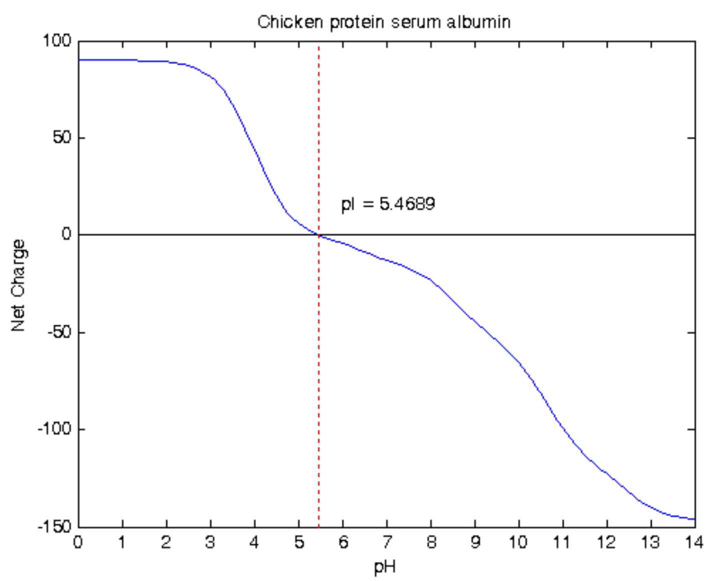
2. You put a raw egg in a vinegar solution with pH 5.

   A. Based on the diagram below, what is the net charge of the protein at neutral pH, (i.e. pH 7)? pH 7 = approximately -10

   B. What is the net charge at pH 5? Why did it change? pH 5 = ~0
   changed b/c vinegar is an acid, i.e. has many H+ ions, the ions will bind to the proteins and change the overall charge, this destabilizes the protein so it denatures more easily and "cooks".

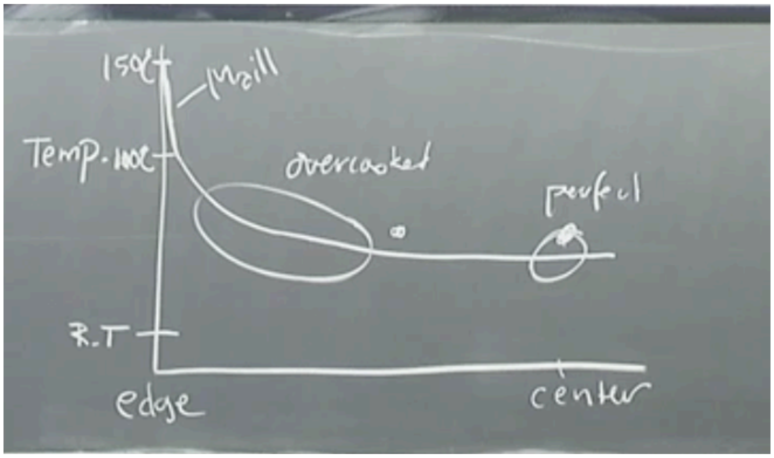
   C. What do you expect to observe when you crack open the egg after marinating in this solution for 1 week? What about if the pH is 3?

   After marinating for a week at pH 5: no or very little change
   At pH 3: will see change in texture



Heat Transfer

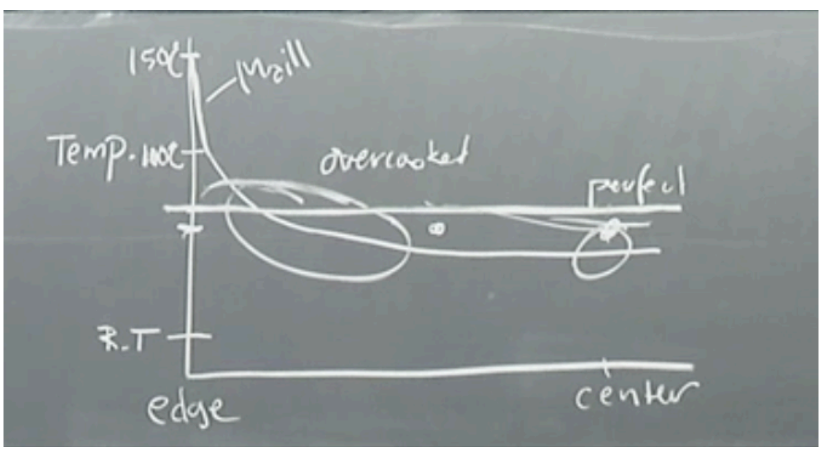
1. Chef Corey seared tuna. It turns out great: perfect temperature at the center and a nice brown crust. Draw a diagram of the temperature in the fish as a function of distance from the center *right after you take* it off the griddle.



2. Explain why it is hard to cook a piece of tuna (or steak).
(1) In order to get a perfect temp center and a brown surface – most of the steak will be overcooked. (2) the steak continues cooking after you take it off the griddle (heat diffuses inwards; not out b/c air is a good thermal insulator) (3) for browning to occur water must evaporate on the surface.
3. The fish has a slightly browned surface. Why? You decide to sprinkle the surface with a tiny bit of sugar. What happens? Why?

Maillard reactions occurred between amino acids and sugars in the meat because of the high heat of the griddle (~150˚C). When you sprinkle sugar, you allow for further Maillard reactions between sugar and amino acids, and also caramelization reactions (=sugar+heat) → the surface gets even browner

4. On your diagram in 1, draw the temperature in the fish as a function of distance from the center *about 15 min after* taking it off the griddle. The straight line (or something much more flat)



Viscosity

1. Rank the following food items below from the least viscous to the most viscous. Explain why you ranked them as you did.
   Water, Cream, honey, peanut butter

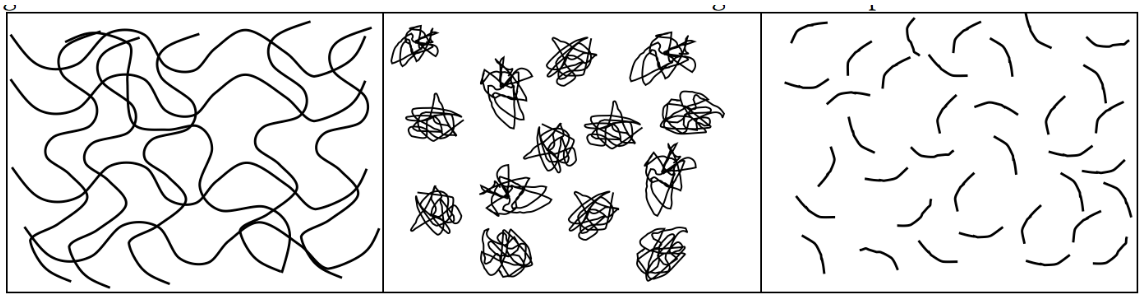
2. Circle one:

Water is a	**simple fluid**	complex fluid
Wine is a	simple fluid	**complex fluid**
Mac&Cheese sauce is a	simple fluid	**complex fluid**

3. Match the blanks. Viscosity increases when kinetic energy of molecules __________; viscosity __________when the volume fraction decreases.

a) decreases/decreases
b) decreases/increases
c) increases/increases
d) increases/decreases

4a. The images below show cartoons of three sauces from a microscopic view. Rank them from expected lowest to highest viscosity.



highest middle lowest

b. What type of thickener would you expect that each image represents? How does it thicken?
   Left: modernist thickener, entangles and slows flow
   Middle: starch, swells and leaks polymers, ie increases volume fraction and slows flow
   Right: can be any material, thickens by slowing flow

Emulsions and Foams

1. You emulsify 30 mL of olive oil into one (10 ml) egg yolk to make aioli. What is occurring at a microscopic scale as you form this emulsion?

We are forming droplets of oil in water, which are stabilized by surfactants (in this case lecithin/phospholipids) in the egg yolks.

2. Why do you need to put in energy to make an emulsion?

Because it requires energy to break up large bubbles into smaller which creates more surface area (i.e. more surface tension/surface energy).

3. What is the volume fraction of the oil in the aioli?

Φ = (30 mL) / (30 mL + 10 mL) = 0.75 = 75%

4. Based on your calculated volume fraction of olive oil in aioli, what is the elastic modulus of this emulsion, assuming σ_{oil} = 10 mN/m and the droplets ~10 µm in diameter?

$E = (\sigma/R)(\phi - \phi_C) = ((10\times10^{-3}\ N/m)/(5\times10^{-6}\ m))\ (0.75-0.64) = 220\ N / m^2$

5. All of these foods are common emulsifiers except:
    - a) egg yolk
    - b) garlic
    - c) mustard
    - d) **brine**